

HI-MAPF — Towards Resource-Efficient Multi-Agent Deployment

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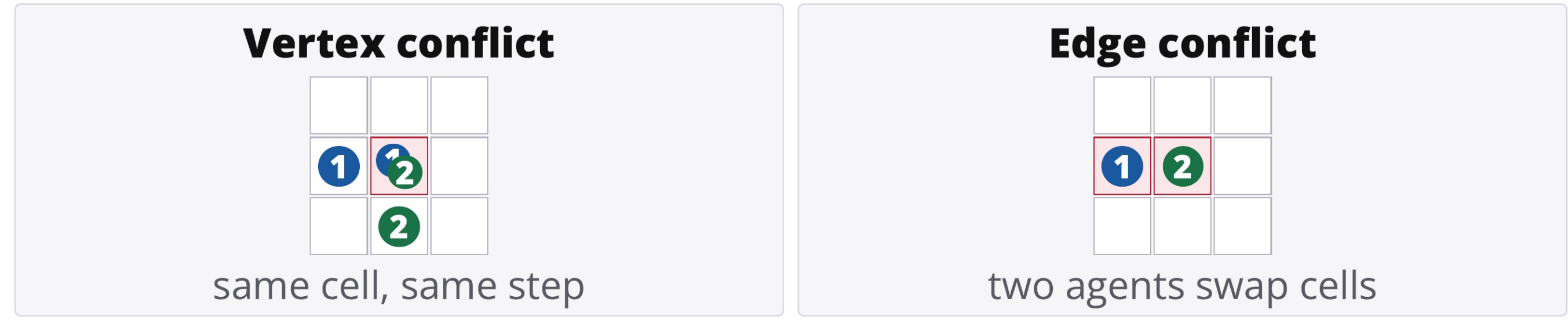


2×–510× fewer Information Units (IU) than learning-based baselines in simulation

The Deployment Bottleneck

PROBLEM 01

MAPF plans collision-free paths for n agents on a graph $G = (V, E)$, avoiding both **vertex** and **edge** conflicts.



THE MISSING QUESTION: WHAT MUST EACH ROBOT KNOW?

Success rate and makespan hide the **sensing, communication and computation** needed to deploy a method. On real robots, **these information demands can limit scalability of deployment**.

PARADIGM	INFORMATION USED → BURDEN
Centralized CBS, LaCAM	Global state: all starts, goals, maps and joint plans. high bandwidth · single point of failure
Distributed SCRIMP, DCC	FOV from broadcast positions; SCRIMP also sends 512-D messages each step. position broadcasts · neural inference
Hybrid HI-MAPF (ours)	Decentralized planning + a coordinator with conflict-triggered alerts. sparse alerts · no onboard FOV sensing

213× fewer IU than SCRIMP on real robots: 199 vs. 42,413

I-MAPF & Information Units

MODEL + METRIC 03

I-MAPF formalizes the information available to each agent i at time t . **Information Units (IU)** make that information countable across search, learning and hybrid methods:

$$I_i(t) = (M_i, S_i, O_i, E_{ij}(t))$$

M, S **Local knowledge:** map structure plus the agent's own start, goal and plan.

O, E **Coordination knowledge:** other-agent observations or sparse event-triggered alerts.

INFORMATION UNIT: METHOD-AGNOSTIC METRIC

$$1 \text{ IU} = (v, s, t, e)$$

Cell v in state s at timestep t . **Every** sensed observation or transmitted message is a multiset of IUs, so maps, trajectories, constraints and neural messages all become countable in *one* unit. A 512-dim message vector costs **512 IU**; one conflict cell costs **1 IU**.

IU is method-agnostic. Lower IU indicates lower bandwidth, energy and onboard hardware burden in deployment.

0 onboard LiDAR or camera sensors used for HI-MAPF coordination in the controlled grid

HI-MAPF: Proposed Approach

FRAMEWORK 04

HI-MAPF keeps planning decentralized. A lightweight coordinator scans submitted paths and sends an alert **only when it predicts a conflict**; the selected agent first attempts a local repair.

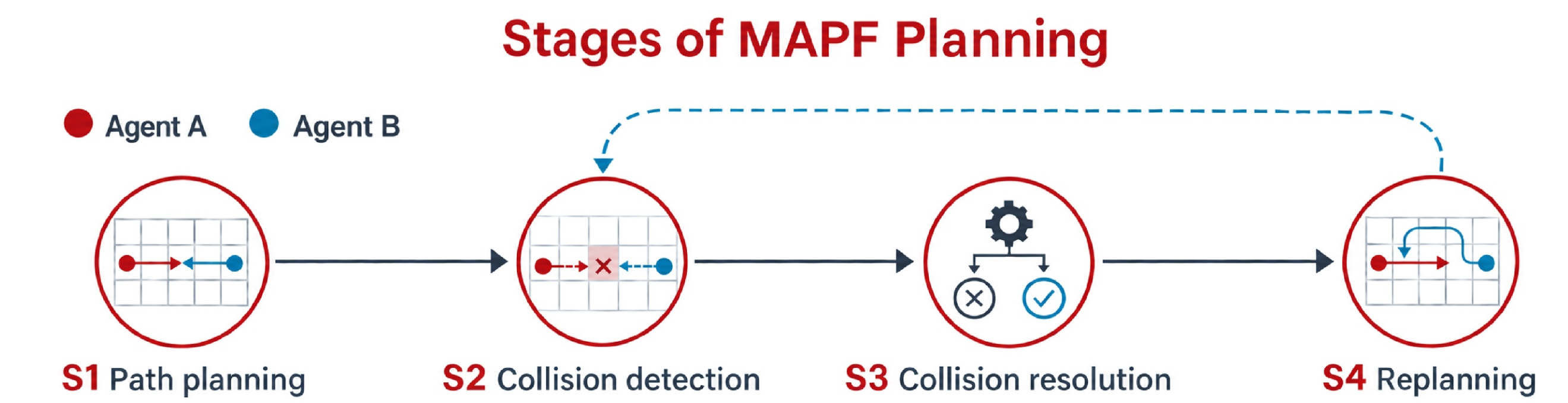


Fig. 1: The detect-control-replan loop, repeated until no conflicts remain.

S1 Plan Plan alone: no other-agent data and no FOV hardware. 0 inter-agent IU	S2 Detect Scan submitted paths for vertex and edge conflicts. paths updated each cycle
S3 Resolve Choose who replans, the rewind window and what to disclose. alert only	S4 Replan Repair locally and resubmit; preserve the safe path segments. Δ_c or r IU

$A(c) = (a_c, t_{j-r}, \Delta_c)$ (alert)

100% hardware success rate, at the lowest makespan of all methods

Tiered Repair: Cheapest Fix First

REPAIR 05

On each conflict, HI-MAPF tries the **least informative repair that works**, escalating only when the cheaper tier fails.

TIER	WHAT IS REVEALED	COST	USE
Yield	Step aside to a parking cell; only cell-occupancy checks.	1 IU	92–99%
Static	Reroute around the conflict cells Δ_c , treated as obstacles.	$ \Delta_c $ IU	1–8%
Dynamic	Short prefixes of the other agents' timed paths.	r IU	<0.3%
Joint planning	Bounded coupled search over the stuck subset.	$ \Delta_c \cdot r$ IU	<0.3%

Targeted replanning. S3 selects the agent with the fewest predicted future collisions.

Bounded disclosure. The alert specifies only who replans, when to rewind and which conflict cells to avoid.

Local recovery. Safe path segments remain unchanged. If all four tiers fail, the agent is deferred (typically at its start) and reintroduced in a second planning phase.

Lower-information repairs first. HI-MAPF increases the information used for coordination only when simpler repairs fail, minimizing average per-conflict communication and computation cost.

Technical Contributions

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FORMULATION

I-MAPF: INFORMATION-CENTRIC FORMULATION

- Models the information available to each agent as $I_i(t)$.
- Includes map knowledge, self-state, other-agent information and event signals.
- Characterizes centralized, distributed, decentralized and hybrid methods by the information available to each agent.

METRIC

IU: INFORMATION UNITS METRIC

- Introduces a method-agnostic metric for information used in coordination.
- Counts atomic facts in maps, observations, paths, constraints and message payloads.
- Compares sensing and communication burden, not the number of messages.

FRAMEWORK

HI-MAPF ALGORITHM

- Combines decentralized planning with centralized conflict detection.
- Sends targeted alerts only when the coordinator predicts a conflict.
- Escalates from local yielding to richer repairs only when simpler tiers fail.

Simulation Results · Four Maps

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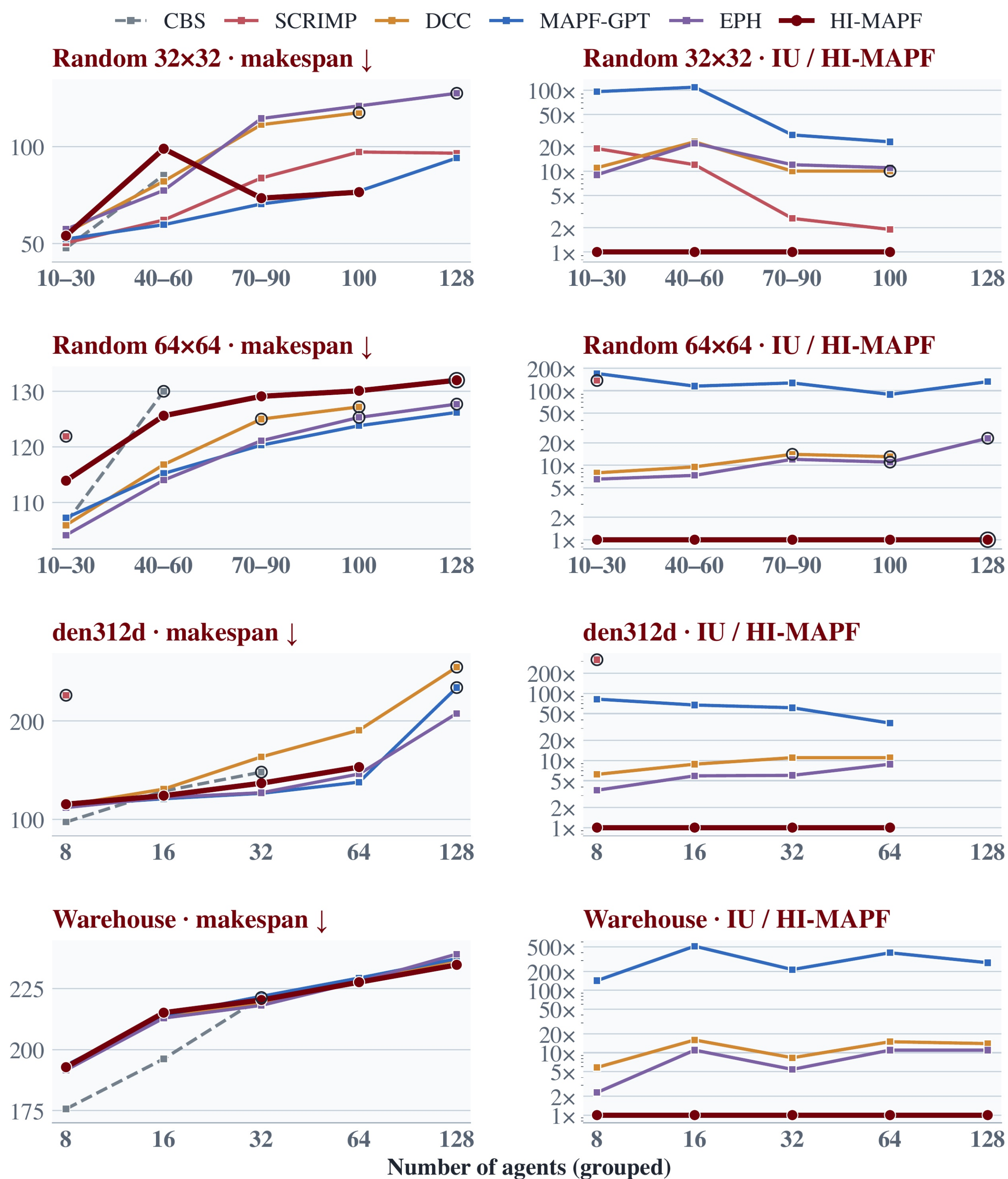


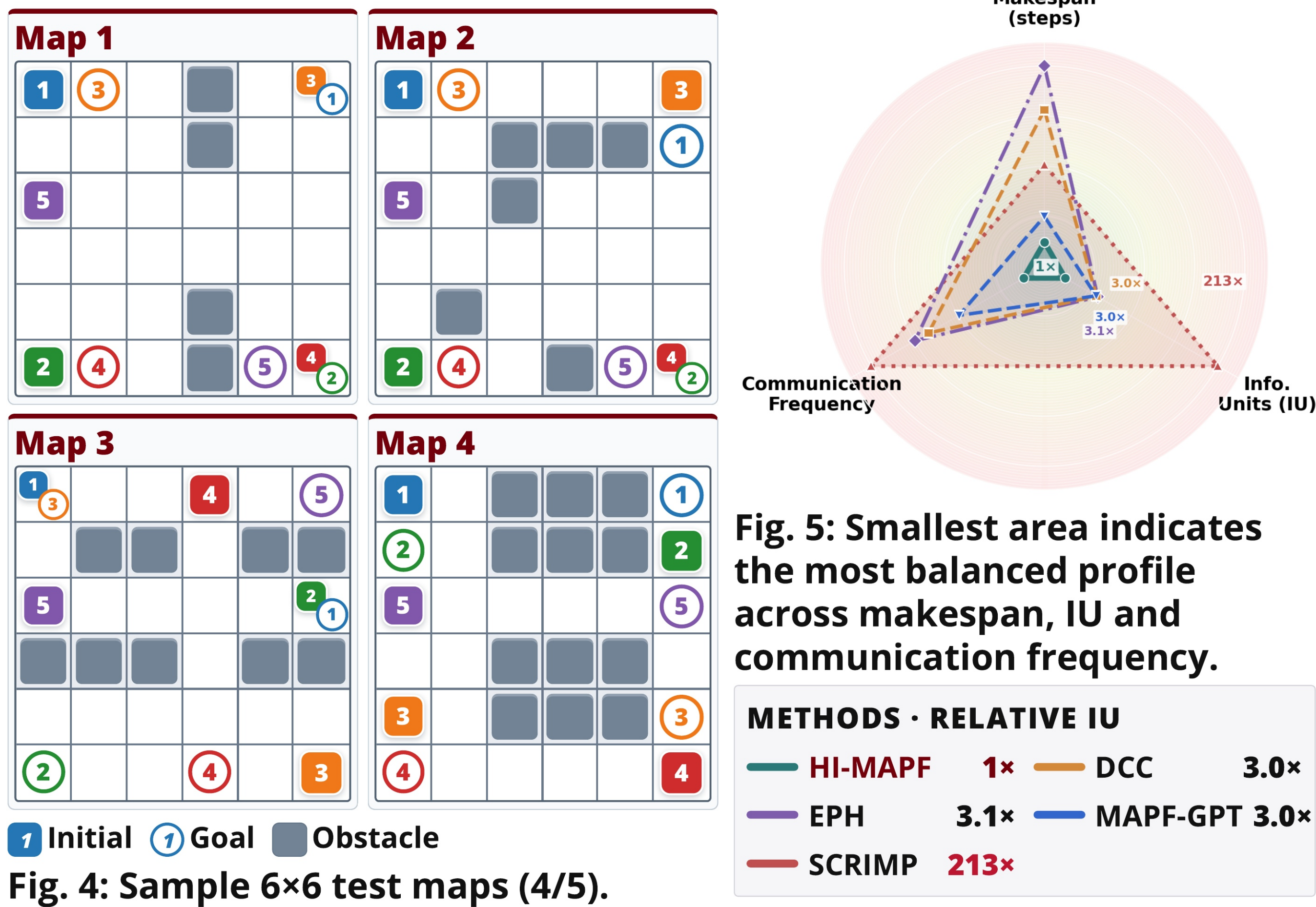
Fig. 2: Four-map simulation results. X-axis: number of agents (grouped); left: makespan (steps); right: IU ratio (baseline ÷ HI-MAPF), with HI-MAPF at 1×. Twenty random instances per setting.

Hardware Evidence

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Fig. 3: Full 6×6 deployment setup with five TurtleBot4 robots.



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What Differentiates HI-MAPF

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EVENT-TRIGGERED INFORMATION SHARING

Event-triggered

Short conflict-specific path fragments are sufficient; no continuous observation or per-timestep messaging is needed.

METHOD-AGNOSTIC COMPARISON

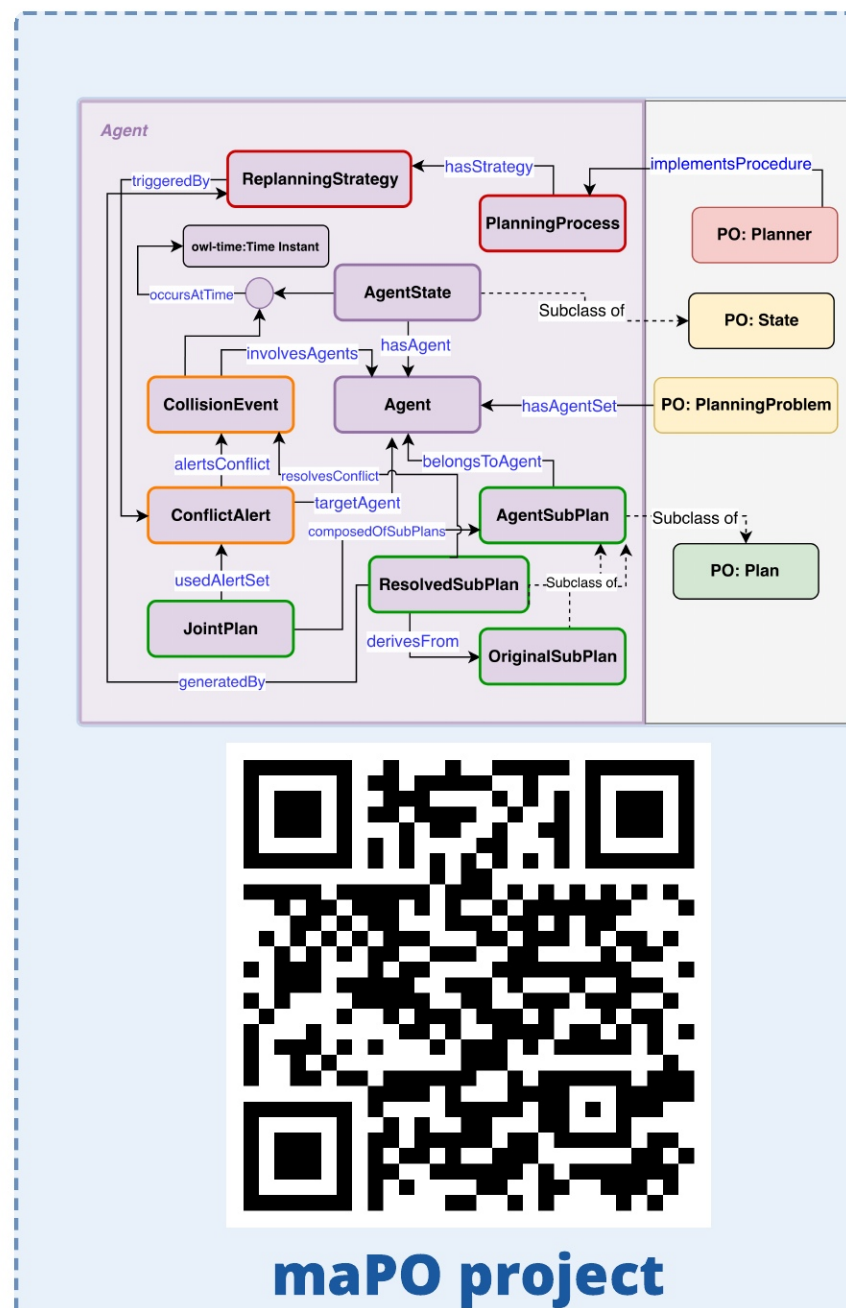
2×–510× fewer IU

Information Units compare information burden across search, learning and hybrid methods.

CONTROLLED-GRID DEPLOYMENT

No LiDAR or camera sensing

HI-MAPF coordinates five TurtleBot4 robots in a controlled 6×6 grid without using onboard LiDAR or camera perception.



maPO: explaining MAPF planner traces

In separate work [6], the **Multi-Agent Planning Ontology** captures repair provenance as linked data: **CollisionEvent** → **ConflictAlert** → **ReplanningStrategy** → **ResolvedSubPlan**.

Reusing W3C **PROV-O** and **OWL-Time**, it connects agents, conflict locations, timestamps and revised subplans.

Trace · explain · audit: Why did an agent yield? Which cells caused the repair? How did its plan change?

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Paper



Website



Video